

PHYSICIAN & DEPARTMENT SQL

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Question :

Perform SQL operations on Physician and Department tables using INNER JOIN, WHERE, LIKE, NOT LIKE, IN, NOT IN, EXISTS, GROUP BY, HAVING, ORDER BY, subqueries, COUNT(), COUNT(DISTINCT), MIN(), and MAX() to retrieve and analyse physician and department information.

Step 1 : Create Physician Database and Table

```
1  -- Step 1: Create Physician Table
2  • CREATE DATABASE Department_db;
3  • USE Department_db;
4  • CREATE TABLE physician (
5      employeeid INT PRIMARY KEY,
6      name VARCHAR(100),
7      position VARCHAR(100),
8      ssn VARCHAR(20)
9  );
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	22:51:44	CREATE DATABASE Department_db	1 row(s) affected
✓ 2	22:51:57	USE Department_db	0 row(s) affected
✓ 3	22:52:02	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected

```

10  -- Step 2: Insert Data into Physician
11  • INSERT INTO physician VALUES
12  (1,'Nikshitha','Staff Internist','111111111'),
13  (2,'Sathwika','Attending Physician','222222222'),
14  (3,'Bhnu Priya','Surgical Attending Physician','333333333'),
15  (4,'Harshika','Senior Attending Physician','444444444'),
16  (5,'Varha','Head Chief of Medicine','555555555'),
17  (6,'Manu','Surgical Attending Physician','666666666'),
18  (7,'Shiny','Surgical Attending Physician','777777777'),
19  (8,'Pranitha','MD Resident','888888888'),
20  (9,'Angela','Attending Psychiatrist','999999999');

```

Output

Action Output

#	Time	Action	Message
✓ 1	22:51:44	CREATE DATABASE Department_db	1 row(s) affected
✓ 2	22:51:57	USE Department_db	0 row(s) affected
✓ 3	22:52:02	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
✓ 4	22:55:20	INSERT INTO physician VALUES (1,'Nikshitha','Staff Internist','111111111'), (2,'Sathwika','Attending Physician','2...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0

```

21  -- Step 3: Create Department Table
22  • CREATE TABLE department (
23      departmentid INT PRIMARY KEY,
24      name VARCHAR(100),
25      head INT
26  );

```

Output

Action Output

#	Time	Action	Message
✓ 1	22:51:44	CREATE DATABASE Department_db	1 row(s) affected
✓ 2	22:51:57	USE Department_db	0 row(s) affected
✓ 3	22:52:02	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
✓ 4	22:55:20	INSERT INTO physician VALUES (1,'Nikshitha','Staff Internist','111111111'), (2,'Sathwika','Attending Physician','2...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓ 5	22:56:02	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected

```

27  -- Step 4: Insert Data into Department
28  • INSERT INTO department VALUES
29  (1,'General Medicine',4),
30  (2,'Surgery',7),
31  (3,'Psychiatry',9);

```

Output

Action Output

#	Time	Action	Message
✓ 1	22:51:44	CREATE DATABASE Department_db	1 row(s) affected
✓ 2	22:51:57	USE Department_db	0 row(s) affected
✓ 3	22:52:02	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
✓ 4	22:55:20	INSERT INTO physician VALUES (1,'Nikshitha','Staff Internist','111111111'), (2,'Sathwika','Attending Physician','2...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓ 5	22:56:02	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓ 6	22:56:52	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Step 2 : JOIN

```

32  -- Step 5: INNER JOIN
33  • SELECT d.name AS Department,
34         p.name AS Head_Physician
35  FROM department d
36  JOIN physician p
37  ON d.head = p.employeeid;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department	Head_Physician
General Medicine	Harshika
Surgery	Shiny
Psychiatry	Angela

Result 1 x

Output

Action Output

#	Time	Action	Message
✓	2	22:51:57	USE Department_db
✓	3	22:52:02	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA... 0 row(s) affected
✓	4	22:55:20	INSERT INTO physician VALUES (1,'Nikshitha','Staff Internist','111111111'), (2,'Sathwika','Attending Physician... 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓	5	22:56:02	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT) 0 row(s) affected
✓	6	22:56:52	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	7	22:57:18	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned

Step 3 : WHERE

```

38  -- Step 6: WHERE Clause
39  • SELECT *
40  FROM physician
41  WHERE position = 'Surgical Attending Physician';
42  -- Step 7: WHERE + LIKE

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	employeeid	name	position	ssn
▶	3	Bhnu Priya	Surgical Attending Physician	333333333
	6	Manu	Surgical Attending Physician	666666666
	7	Shiny	Surgical Attending Physician	777777777
•	NULL	NULL	NULL	NULL

physician 2 x

Output

Action Output

#	Time	Action	Message
✓	3	22:52:02	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA... 0 row(s) affected
✓	4	22:55:20	INSERT INTO physician VALUES (1,'Nikshitha','Staff Internist','111111111'), (2,'Sathwika','Attending Physician... 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓	5	22:56:02	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT) 0 row(s) affected
✓	6	22:56:52	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	7	22:57:18	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned
✓	8	22:57:59	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned

Step 4 : LIKE

```

42  -- Step 7: WHERE + LIKE
43  • SELECT *
44  FROM physician
45  WHERE name LIKE 'John%';

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	employeeid	name	position	ssn
▶	3	Bhnau Priya	Surgical Attending Physician	333333333
	6	Manu	Surgical Attending Physician	666666666
	7	Shiny	Surgical Attending Physician	777777777
•	NULL	NULL	NULL	NULL

physician 3 x

Output

Action Output

#	Time	Action	Message
✓ 4	22:55:20	INSERT INTO physician VALUES (1,'Nikshitha','Staff Internist','111111111'), (2,'Sathwika','Attending Physician'...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓ 5	22:56:02	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓ 6	22:56:52	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	22:57:18	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 8	22:57:59	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 9	22:58:46	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned

Step 5 : COUNT

```

46  -- Step 8: COUNT()
47  • SELECT COUNT(*) AS Total_Physicians
48  FROM physician;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Physicians
▶ 9

Result 4 x

Output

Action Output

#	Time	Action	Message
✓ 5	22:56:02	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓ 6	22:56:52	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	22:57:18	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 8	22:57:59	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 9	22:58:46	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 10	22:59:26	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned

```

49  -- Step 9: COUNT DISTINCT
50  • SELECT COUNT(DISTINCT position) AS Different_Positions
51  FROM physician;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

Different_Positions
7

Result 5 x

Output

Action Output

#	Time	Action	Message
✓ 6	22:56:52	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	22:57:18	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 8	22:57:59	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 9	22:58:46	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 10	22:59:26	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	23:00:04	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned

Step 6 : GROUP BY

```

52  -- Step 10: GROUP BY
53  • SELECT position,
54        COUNT(*) AS Total_Employees
55  FROM physician
56  GROUP BY position;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 6 x

Output

Action Output

#	Time	Action	Message
✓ 7	22:57:18	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 8	22:57:59	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 9	22:58:46	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 10	22:59:26	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	23:00:04	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 12	23:00:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned

```

57  -- Step 11: GROUP BY + HAVING
58  •  SELECT position,
59         COUNT(*) AS Total_Employees
60  FROM physician
61  GROUP BY position
62  HAVING COUNT(*) > 1;

```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	position	Total_Employees
▶	Surgical Attending Physician	3

Result 7 x

Output

 Action Output ▼

	#	Time	Action	Message
✓	8	22:57:59	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓	9	22:58:46	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓	10	22:59:26	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓	11	23:00:04	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓	12	23:00:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓	13	23:01:20	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

Step 7: ORDER BY

```

63 -- Step 12: ORDER BY ASC
64 • SELECT *
65 FROM physician
66 ORDER BY name ASC;

```

Result Grid				
Filter Rows:				
Edit:				
Export/Import:				
Wrap Cell Content:				
	employeeid	name	position	ssn
▶	9	Angela	Attending Psychiatrist	999999999
	3	Bhnau Priya	Surgical Attending Physician	333333333
	4	Harshika	Senior Attending Physician	444444444
	6	Manu	Surgical Attending Physician	666666666
	1	Nikshitha	Staff Internist	111111111
	8	Pranitha	MD Resident	888888888
	2	Sathwika	Attending Physician	222222222
	7	Shiny	Surgical Attending Physician	777777777
	5	Varha	Head Chief of Medicine	555555555
•	NULL	NULL	NULL	NULL

physician 8 ×

Output				
Action Output				
#	Time	Action	Message	
✓ 9	22:58:46	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned	
✓ 10	22:59:26	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned	
✓ 11	23:00:04	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned	
✓ 12	23:00:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned	
✓ 13	23:01:20	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned	
✓ 14	23:02:15	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned	


```

67  -- Step 13: ORDER BY DESC
68  •  SELECT *
69  FROM physician
70  ORDER BY employeeid DESC;

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
employeeid	name	position	ssn	
9	Angela	Attending Psychiatrist	999999999	
8	Pranitha	MD Resident	888888888	
7	Shiny	Surgical Attending Physician	777777777	
6	Manu	Surgical Attending Physician	666666666	
5	Varha	Head Chief of Medicine	555555555	
4	Harshika	Senior Attending Physician	444444444	
3	Bhna Priya	Surgical Attending Physician	333333333	
2	Sathwika	Attending Physician	222222222	
1	Nikshitha	Staff Internist	111111111	
NULL	NULL	NULL	NULL	

physician 9 x

Output

Action Output

#	Time	Action	Message
✓ 10	22:59:26	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	23:00:04	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 12	23:00:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 13	23:01:20	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 14	23:02:15	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 15	23:02:41	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned

Step 8 : IN


```

71 -- Step 14: WHERE + IN
72 • SELECT *
73 FROM physician
74 WHERE employeeid IN
75 (
76     SELECT head
77     FROM department
78 );

```

Result Grid				
Filter Rows:				
Edit: Export/Import: Wrap Cell Content:				
	employeeid	name	position	ssn
▶	4	Harshika	Senior Attending Physician	444444444
	7	Shiny	Surgical Attending Physician	777777777
	9	Angela	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 10 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 11	23:00:04	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned	
✓ 12	23:00:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned	
✓ 13	23:01:20	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned	
✓ 14	23:02:15	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned	
✓ 15	23:02:41	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned	
✓ 16	23:03:19	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned	

Step 9 : NOT IN

```

79  -- Step 15: NOT IN
80  • SELECT *
81  FROM physician
82  WHERE employeeid NOT IN
83  (
84      SELECT head
85      FROM department
86  );

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
employeeid	name	position	ssn	
1	Nikshitha	Staff Internist	111111111	
2	Sathwika	Attending Physician	222222222	
3	Bhna Priya	Surgical Attending Physician	333333333	
5	Varha	Head Chief of Medicine	555555555	
6	Manu	Surgical Attending Physician	666666666	
8	Pranitha	MD Resident	888888888	
*	NULL	NULL	NULL	

physician 11 x

Output

#	Time	Action	Message
12	23:00:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	23:01:20	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
14	23:02:15	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
15	23:02:41	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
16	23:03:19	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
17	23:04:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned

Step 10 : GROUP BY + ORDER BY

```

87  -- Step 16: GROUP BY + ORDER BY
88  •  SELECT position,
89      COUNT(*) AS Total_Employees
90  FROM physician
91  GROUP BY position
92  ORDER BY Total_Employees DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 12 x

Output

Action Output

#	Time	Action	Message
✓ 13	23:01:20	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 14	23:02:15	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 15	23:02:41	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 16	23:03:19	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 17	23:04:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 18	23:06:24	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned

Step 11: LIKE

```

93  -- Step 17: WHERE + LIKE
94  •  SELECT *
95  FROM physician
96  WHERE position LIKE '%Attending%';

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 13 x

Output

Action Output

#	Time	Action	Message
✓ 14	23:02:15	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 15	23:02:41	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 16	23:03:19	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 17	23:04:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 18	23:06:24	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 19	23:07:17	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned

Step 12: JOIN + WHERE

```

97 -- Step 18: JOIN + WHERE
98 • SELECT p.name
99 FROM physician p
100 JOIN department d
101 ON p.employeeid = d.head
102 WHERE d.name = 'Surgery';

```

Result Grid Filter Rows: | Export: | Wrap Cell Content:

	name
▶	Shiny

Result 14 x

Output

Action Output

#	Time	Action	Message
✓ 15	23:02:41	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 16	23:03:19	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 17	23:04:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 18	23:06:24	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 19	23:07:17	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 20	23:08:12	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned

Step 13: JOIN+ORDER BY

```

103 -- Step 19: JOIN + ORDER BY
104 • SELECT d.name AS Department,
105        p.name AS Head_Physician
106 FROM department d
107 JOIN physician p
108 ON d.head = p.employeeid
109 ORDER BY d.name;

```

Result Grid Filter Rows: | Export: | Wrap Cell Content:

	Department	Head_Physician
▶	General Medicine	Harshika
	Psychiatry	Angela
	Surgery	Shiny

Result 15 x

Output

Action Output

#	Time	Action	Message
✓ 16	23:03:19	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 17	23:04:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 18	23:06:24	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 19	23:07:17	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 20	23:08:12	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 21	23:09:02	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned

Step 14: NOT LIKE

```

110 -- Step 20: WHERE + NOT LIKE
111 • SELECT *
112 FROM physician
113 WHERE position NOT LIKE '%Surgical%';

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [IA](#)

	employeeid	name	position	ssn
▶	1	Nikshitha	Staff Internist	111111111
	2	Sathwika	Attending Physician	222222222
	4	Harshika	Senior Attending Physician	444444444
	5	Varha	Head Chief of Medicine	555555555
	8	Pranitha	MD Resident	888888888
	9	Angela	Attending Psychiatrist	999999999
•	NULL	NULL	NULL	NULL

physician 16 x

Output

Action Output

#	Time	Action	Message
✓ 17	23:04:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 18	23:06:24	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 19	23:07:17	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 20	23:08:12	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 21	23:09:02	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 22	23:09:45	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned

Step 15: COUNT+HAVING

```

114 -- Step 21: COUNT + HAVING
115 • SELECT position,
116         COUNT(*) AS Total
117 FROM physician
118 GROUP BY position
119 HAVING COUNT(*) >= 2;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	position	Total
▶	Surgical Attending Physician	3

Result 17 x

Output

Action Output

#	Time	Action	Message
✓ 18	23:06:24	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 19	23:07:17	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 20	23:08:12	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 21	23:09:02	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 22	23:09:45	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓ 23	23:10:29	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned

Step 16: EXISTS

```
120  -- Step 22: EXISTS
121  • SELECT *
122  FROM department d
123  WHERE EXISTS
124  (
125      SELECT 1
126      FROM physician p
127      WHERE p.employeeid = d.head
128  );
```

Result Grid

	departmentid	name	head
1	1	General Medicine	4
2	2	Surgery	7
3	3	Psychiatry	9
4	NULL	NULL	NULL

department 18 x

Output

Action Output

#	Time	Action	Message
19	23:07:17	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
20	23:08:12	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
21	23:09:02	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
22	23:09:45	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
23	23:10:29	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
24	23:11:17	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned

Step 17: MAX

```
129  -- Step 23: MAX()
130  • SELECT MAX(employeeid) AS Highest_Employee_ID
131  FROM physician;
```

Result Grid

	Highest_Employee_ID
1	9

Result 19 x

Output

Action Output

#	Time	Action	Message
20	23:08:12	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
21	23:09:02	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
22	23:09:45	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
23	23:10:29	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
24	23:11:17	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned
25	23:11:58	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned

Step 18: MIN

```

132 -- Step 24: MIN()
133 • SELECT MIN(employeeid) AS Lowest_Employee_ID
134 FROM physician;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Lowest_Employee_ID			
▶ 1			

Result 20 x

Output

Action Output

#	Time	Action	Message
✓ 21	23:09:02	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 22	23:09:45	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓ 23	23:10:29	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
✓ 24	23:11:17	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned
✓ 25	23:11:58	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 26	23:12:49	SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned